

$$\square = 39$$

3 D

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi\left(\frac{d}{2}\right)^3}$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + 3\left(\frac{\Delta d}{d}\right)$$

$$\frac{\Delta\rho}{\rho} \times 100\% = \frac{\Delta M}{M} \times 100\% + 3\left(\frac{\Delta d}{d}\right) \times 100\%$$

$$= 1 + 3(3) = 10\%$$

4 C $\epsilon = 1/6 \sigma t^2$